Berger–Hopf Standard Model: Action-domain localization and an auditable route to physical predictions

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Abstract

The Berger–Hopf Standard Model (BHSM) investigates whether a common stratified variational framework can organize Standard Model-associated structure without using measured masses or mixing angles to choose its upstream dynamics. We formulate the current evidence boundary around action-domain localization, finite-history response, and the transfer from certified mathematical objects to physical observables. The authorized AE3 extension promotes a retained reciprocal-join profile to a response constraint, yielding a regular local material carrier without a new continuous coefficient. Separately, a 512-bit interval calculation composes the frozen central scalar response across 370 intervals and 371 nodes, with maximum response-norm upper bound 8.405509181456809. We describe the complete ambient midpoint chain rule required for the remaining signed transverse calculation and an explicit bound on stored coordinate-solve error. These results do not establish the joint physical background, particle masses, mixing matrices, scattering observables, or full BHSM completion. The manuscript binds numerical values to immutable source artifacts and identifies the remaining proof operands that must precede a physical prediction.

1 Scientific scope and provenance

BHSM is an evolving mathematical and computational research program. Its historical spectral and coupling screens, conditional structural results, numerical center calculations, and outward certificates have different logical authority. A numerical match cannot close a missing action, domain, normalization, or observable map. The present draft therefore organizes results by their actual hypotheses and reproducible evidence, rather than treating an earlier release label as physical completion.

The current definition of done requires every claimed quantity to share one declared parent action and input ledger, with the intervening variational, operator, scale, and observable maps closed [2, 3]. Measured mass and mixing data may enter a later comparison; they may not select a branch, coefficient, normalization, regularization, or eigenvector ordering upstream. Frozen historical outputs remain unchanged and retain their original labels.

This manuscript is the working source for the current flagship synthesis. It reports completed local results and a precisely delimited computational frontier. It is not a submission-ready claim of a complete physical model.

2 Stratified action and physical ownership

The retained architecture distinguishes a provisional S^8 parent, a relative two-cap $S^{5|4}$ construction, and an intrinsic localized four-dimensional effective theory. Compatibility maps between these strata are part of the model specification. Fields, coefficients, and domains cannot be moved between strata without an explicit action-owned map.

At the current finite-history frontier, geometry-local derivative kernels are necessary computational inputs. They are not by themselves the complete gauge, ghost, fermion, scalar, and mixed-field derivatives of one physical background. Similarly, an implemented spectrum, vertex, or collision engine does not supply the action-derived operands that it is designed to consume. These distinctions prevent a software capability from being counted as a physical prediction [2, 3].

The current program combines compatible local pieces as they become available. A composed claim is permitted only when its pieces share the action version, background, domain, scale and renormalization conventions, and provenance chain. Later candidate action representatives are not silently identified with an earlier certified representative.

3 The AE3 local material carrier

3.1 Response constraint

The retained reciprocal-join profile is

$$\sigma_0(\chi) = -\frac{1}{2} + \frac{2\chi}{\pi} - \frac{\sin(4\chi)}{2\pi}, \qquad 0 \le \chi \le \frac{\pi}{2}. \tag{1}$$

AE3 promotes this previously fixed profile into the existing eta-to-sigma response constraint. On the oriented orbit space Q, define

$$W_J[\eta] = \sin^2 f \cos^2 f, \qquad Z_J[\eta] = \int_Q W_J[\eta]\, d\ell, \tag{2}$$

and impose

$$C_\sigma = n_\eta^A \nabla_A \sigma - \frac{W_J[\eta]}{Z_J[\eta]} = 0, \qquad \sigma|_{Q_-} = -\frac{1}{2}. \tag{3}$$

The nonpropagating multiplier λ_σ supplies the response term

$$S_{\text{response}} = \int dt \int_Q d\ell\, \mu_Q \lambda_\sigma C_\sigma. \tag{4}$$

The corresponding action replaces the old fixed profile wherever it occurs and adds this constraint once. It does not add the retained Hopf/FR energy a second time. This authorized action-domain promotion introduces no new continuous coefficient, physical scale, or particle label [4].

3.2 Regularity and scope

On the retained identity branch,

$$\sigma_0'(\chi) = \frac{4}{\pi}\sin^2(2\chi) > 0 \quad (0 < \chi < \pi/2), \qquad \sigma_0(\pi/4) = 0, \quad \sigma_0'(\pi/4) = \frac{4}{\pi}. \tag{5}$$

Thus the material zero is transverse. The local carrier is

$$D_{\text{enc}} = \{\sigma < 0\}, \qquad \Sigma_{\text{enc}} = \{\sigma = 0\}, \qquad n = \frac{\nabla\sigma}{|\nabla\sigma|}. \tag{6}$$

The retained weight $\Lambda(\sigma) = 1 - 4\sigma^2$ is the unique even polynomial of degree at most two with unit value at the seam and zeros at $\sigma = \pm 1/2$. The associated boundary and flux cancellation statements refer to this smooth internal material interface. The event-time condition and the material surface are distinct objects.

The current algebraic transport retains three charged sectors and three family slots. The resulting nine fibers preserve their supplied BHSM state labels through the carrier construction; the geometry does not independently select a particle species. This does not establish commutation with the still-missing interacting history Hamiltonian or a persistent full-field particle [2]. The older unchanged-AE2 audit, in which none of six candidate carriers qualified, remains a valid historical result within AE2.

4 Complete finite-history second variation

4.1 Why the midpoint requires the full ambient space

Let E be the 99-by-74 augmented trial frame, e its current-Green coordinate axis, and V the stored 74-by-73 transverse basis. The retained midpoint tensor contains $H[U, U]$, where $U = EV$ and $H = D^2F$. Transverse endpoint variations need not remain transverse at a Hermite–Simpson midpoint. They can acquire axis and frame-normal components. Dropping those terms is not justified by a small binary64 residual.

For fixed interval length h_i, the endpoint-pair midpoint map has blocks

$$M_L = \tfrac{1}{2} E_i P_i + \tfrac{h_i}{8} DF_i E_i P_i, \tag{7}$$

$$M_R = \tfrac{1}{2} E_{i+1} P_{i+1} - \tfrac{h_i}{8} DF_{i+1} E_{i+1} P_{i+1}. \tag{8}$$

The fixed reset has zero left variation at interval zero. Complete the actual retained directions to a square ambient basis $S = [U\ C]$, with 26 complement columns, and solve

$$M = [M_L\ M_R] = UA + CD. \tag{9}$$

For each output component, the full pullback is

$$H[M, M] = A^T Q_{UU} A + A^T Q_{UC} D + D^T Q_{UC}^T A + D^T Q_{CC} D. \tag{10}$$

Both cross terms and the complement-complement block are required. Their signs may reinforce or cancel the retained contribution. An incomplete transverse-only norm is neither an upper nor a lower bound for the full pullback [5].

4.2 Signed assembly before norms

Writing $\mathcal{A}_i = -R_i^{-1} E_{i+1}^T$ for the unchanged reduced solve and test-frame map, the local second response is

$$\begin{aligned} \mathcal{S}_i = &\left(\frac{h_i}{6} \mathcal{A}_i + \frac{h_i^2}{12} \mathcal{A}_i DF_{i+1/2} \right) B_i \\ &+ \frac{2h_i}{3} \mathcal{A}_i H_{i+1/2}[M_i, M_i] \\ &+ \left(\frac{h_i}{6} \mathcal{A}_i - \frac{h_i^2}{12} \mathcal{A}_i DF_{i+1/2} \right) B_{i+1}. \end{aligned} \tag{11}$$

All tensors are first pulled back to the common endpoint-pair coordinates. The midpoint incidence terms retain their signs. If a local block is T and a causal transport is P, the identity

$$\|PT\|_F^2 = \operatorname{tr}(P, TT^T P^T) \tag{12}$$

retains output covariance through the transport. Contributions from adjacent intervals acting on the same global diagonal input block must also be added with their signed cross covariance before taking a norm. The existing block-sup proof domain supplies the remaining triangle inequalities.

4.3 Stored-basis and coordinate-solve verification

Binary64 center assembly is distinct from an outward enclosure. For an exact stored binary64 basis S, a 512-bit Arb inverse enclosure verifies nonsingularity without deleting a small singular direction. If $\widehat{X}$ is a computed coordinate matrix and M the exact stored binary64 target,

$$\|S^{-1}M - \widehat{X}\|_\infty \leq \|S^{-1}\|_\infty \, \|M - S\widehat{X}\|_\infty. \tag{13}$$

Arb supplies the residual and inverse bounds [8, 7]. Equation (13) controls this stored solve only. Physical direction construction, Hessian rounding, propagation of coordinate error through (10), and the action-neighborhood remainder are separate required operands.

5 Verified numerical evidence

The completed correlated scalar calculation uses 512-bit interval arithmetic and the frozen causal frames and preconditioner. Table 1 and Figure 1 are generated from the source certificate after its NPZ data hash is verified [6]. No values are fitted to experimental particle data.

Table 1: Completed frozen central scalar calculation. These are scoped mathematical bounds, not physical mass or scattering observables.

Quantity	Recorded result
Intervals / nodes	370 / 371
Arithmetic precision	512 bits
Maximum causal response norm upper	8.405509181456809
Node owning the maximum	370
Maximum local second-residual norm upper	0.08778167488957692

The signed tensor recovery requires 370 endpoints and 370 midpoints. At each midpoint the 26 complement directions require $73 \cdot 26 + 26 \cdot 27/2 = 2249$ new directional pairs, in addition to the retained symmetric transverse block. These are contraction counts, not empirical CPU estimates. At this draft's evidence boundary, the recovery and supplemental campaign are still in progress; no complete signed center coefficient or outward two-radius certificate is reported here.

6 Physical observables and remaining obligations

The project's physical finish line is stronger than successful local certificates. It requires the complete joint finite-history operator oracle, projected heat-minus-zeta force, KKT root, and constrained physical Hessian, together with compatible downstream maps and continuum control. A positive center screen alone cannot establish these objects.

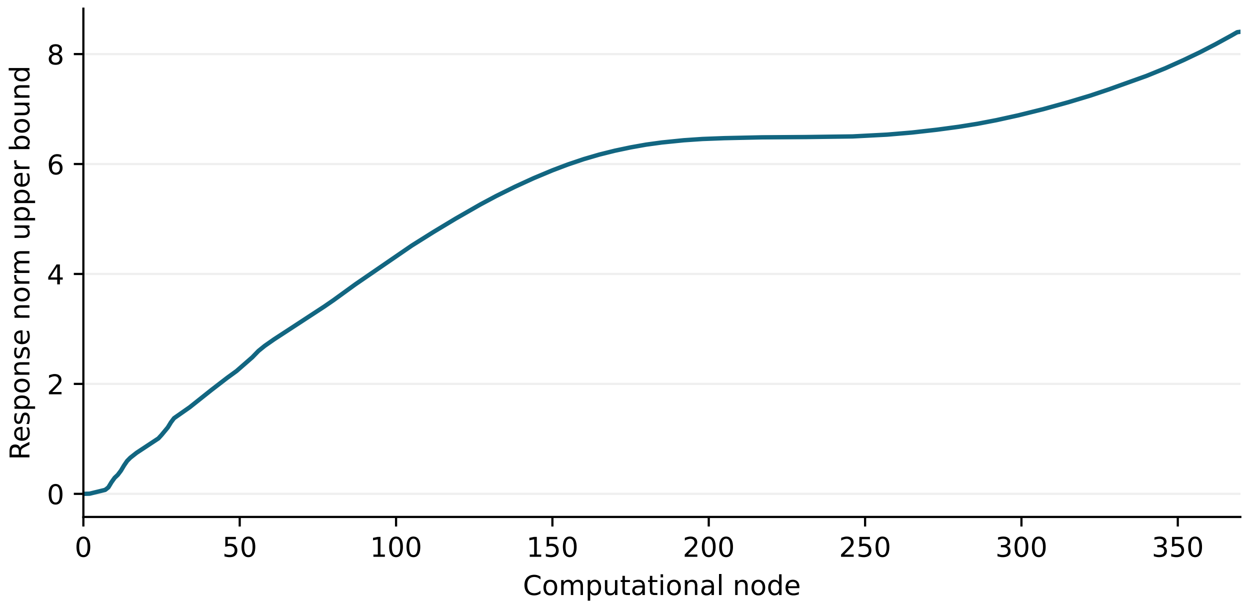

Figure 1: Actual certified numerical response along the frozen path. The horizontal coordinate is a computational node index, not laboratory time. The vertical value is a response-norm upper bound. The central scalar certificate does not enclose the complete transverse neighborhood.

Every claimed observable must eventually be classified once as a derived point prediction, interval prediction, selection rule, qualitative structure, comparison only, open internal blocker, or outside BHSM 1.0 scope [3]. The required coverage includes the universal electromagnetic vertex and muon magnetic moment; stability and decays; spectral enclosures or exclusions; the known-particle inventory; and representative QED, weak, strong, scalar, flavor, and neutrino scattering benchmarks. This draft does not substitute a list of implemented engines for that completed matrix.

The minimal Standard Model and any effective neutrino extension must remain distinguished. No confinement theorem, completed no-extra-light-state theorem, or first-principles derivation of all Standard Model data is asserted. A failed numerical representation screen also does not prove physical instability or nonexistence of a root.

7 Reproduction, data availability, and manuscript status

The plotted values and the table are generated from the immutable scalar source at revision `d75e77bb`; the downloadable Museum dataset records the full 40-character revision and source SHA-256 digests. The exporter refuses an unvalidated certificate, a changed data hash, a mismatch with the recorded source revision, or an inconsistent array shape or maximum. The plotting generator consumes the same dataset. Repeated artifact materializations are compared byte for byte.

The complete scientific source and claim ledgers are available in the repository [1]. This draft is maintained separately from frozen historical manuscripts. Its abstract, results, and conclusions must be updated from completed certificates before external submission; pending physical quantities are not represented by simulated replacement values.

Computational and editorial work used Codex assistance. The scientific provenance is the repository's retained action, derivations, tests, and certificates. Author declarations, venue-specific formatting, and the final scientific review remain part of submission preparation. No journal submission or acceptance is claimed.

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